

Solar Cooker Design Comparison: Types, Principles, and Schematics

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1 Overview of Solar Cooker Types

Table 1: Comparison of main solar cooker types

Type	Max Temp	Efficiency	Cost	Tracking	Best For
Box	150°C	25–35%	\$\$	None	Baking, stews
Parabolic	370°C	50–65%	\$\$\$	Frequent	Frying, grilling
Panel	150°C	20–30%	\$	Occasional	Portable, DIY
Evacuated Tube	280°C	40–50%	\$\$\$\$	None	Cold climates
Hybrid + Storage	150°C	35–55%	\$\$\$\$	Varies	Evening cooking

2 Type 1: Box Solar Cooker

2.1 Working Principle

An insulated box with a glass top creates a greenhouse effect. Sunlight enters through the glass, is absorbed by the black interior, and the trapped heat cooks food slowly at moderate temperatures.

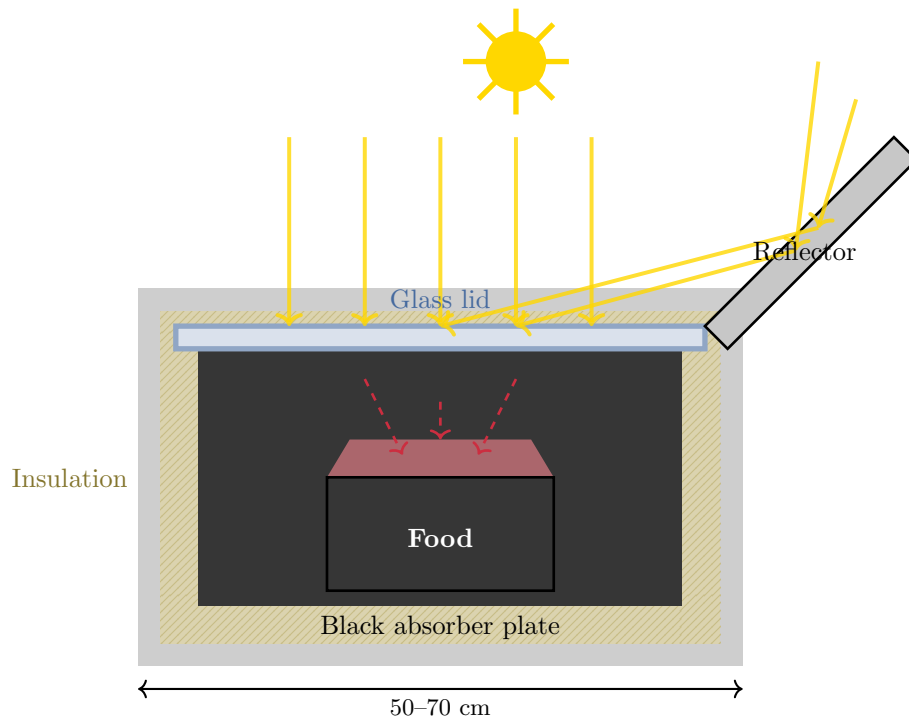


Figure 1: **Box Solar Cooker** – Cross-section showing insulated chamber, glass lid, reflector, and cooking pot. Greenhouse effect traps heat inside.

2.2 Key Features

- Double or triple glass glazing reduces heat loss
- Black interior maximizes absorption (absorptivity $\alpha > 0.95$)
- Insulation (fiberglass, straw, or foam) minimizes conduction losses
- Reflector lid increases solar input by 30–50%
- No tracking required; works for 4–6 hours without adjustment

3 Type 2: Parabolic Solar Cooker

3.1 Working Principle

A parabolic reflector concentrates sunlight to a focal point where the cooking pot is placed. The concentrated rays can achieve very high temperatures for frying and grilling.

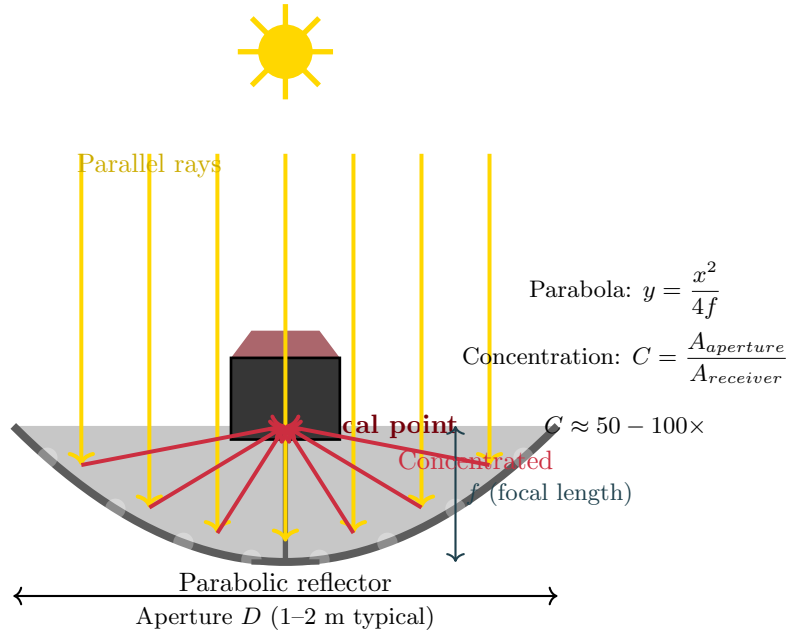


Figure 2: **Parabolic Solar Cooker** – Parallel sun rays are reflected to a focal point where the pot is placed. High concentration ratio enables temperatures $>300^{\circ}\text{C}$.

3.2 Key Features

- Concentration ratio 50–100 $\times$ achieves 300–400 $^{\circ}\text{C}$
- Optimal focal length $f = D/4$ for maximum angle tolerance
- Requires tracking every 15–30 minutes
- Deep dish designs (focus inside dish) are safer
- Can fry, grill, and boil water in minutes

4 Type 3: Panel Solar Cooker (CooKit Style)

4.1 Working Principle

Multiple flat reflective panels direct sunlight onto a cooking pot, which is often placed inside a transparent heat-resistant bag or under an inverted glass bowl to create a mini-greenhouse.

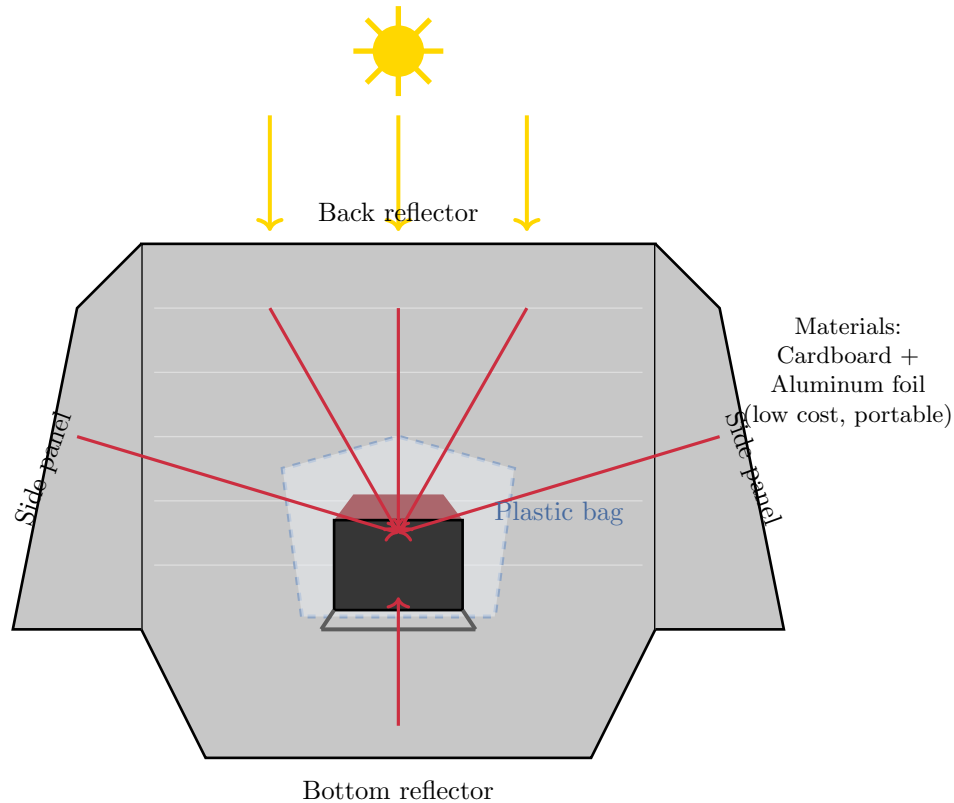


Figure 3: **Panel Solar Cooker (CooKit)** – Flat reflective panels surround a pot in a transparent bag. Simple, cheap, and highly portable.

4.2 Key Features

- Made from cardboard and aluminum foil (\$5–20)
- Folds flat for transport; weighs <1 kg
- Transparent bag/bowl creates greenhouse effect around pot
- Temperatures reach 120–150°C
- Requires repositioning every 1–2 hours
- Ideal for slow cooking rice, beans, stews

5 Type 4: Evacuated Tube Solar Cooker

5.1 Working Principle

Vacuum-insulated glass tubes absorb solar radiation with minimal heat loss. Heat is transferred to a cooking chamber via heat pipes or circulating fluid. Works even in cold weather.

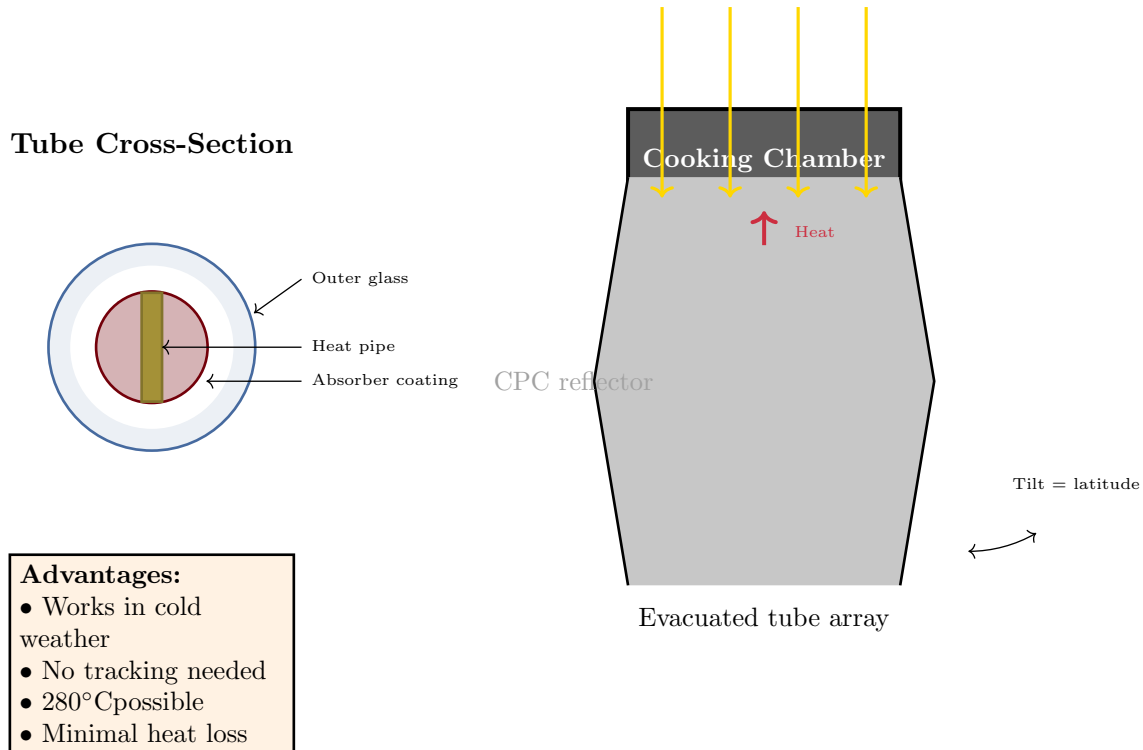


Figure 4: **Evacuated Tube Solar Cooker** – Vacuum insulation minimizes heat loss. Heat pipes transfer energy to cooking chamber. Works in cloudy/cold conditions.

5.2 Key Features

- Vacuum between glass layers eliminates convection/conduction losses
- Selective absorber coating: high absorptivity ($\alpha > 0.92$), low emissivity ($\varepsilon < 0.08$)
- Compound Parabolic Concentrator (CPC) reflector boosts collection
- Can reach 280°C; works even below freezing
- Higher cost but excellent performance in marginal conditions

6 Type 5: Hybrid/Indirect Cooker with Thermal Storage

6.1 Working Principle

Solar collector is separate from cooking area. Heat transfer fluid (oil or water) or phase change material (PCM) stores thermal energy for cooking after sunset or indoors.

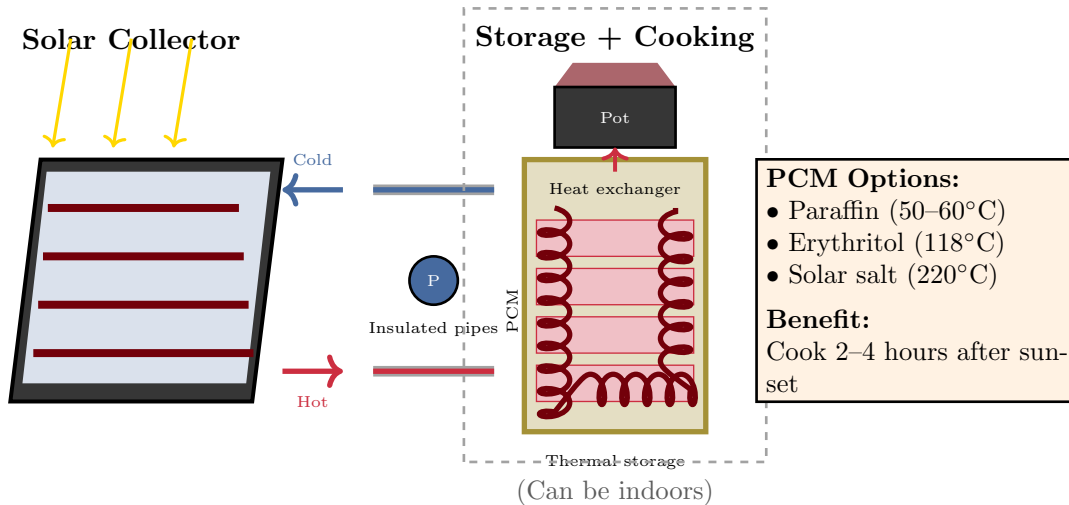


Figure 5: **Hybrid Solar Cooker with Thermal Storage** – Separate collector and cooking unit connected by heat transfer loop. PCM stores heat for evening cooking.

6.2 Key Features

- Cooking can happen indoors, away from collector
- Phase Change Materials (PCM) store latent heat
- Extends cooking 2–4 hours after sunset
- Higher complexity and cost, but most versatile
- Best for community kitchens and consistent daily use

7 Design Comparison Summary

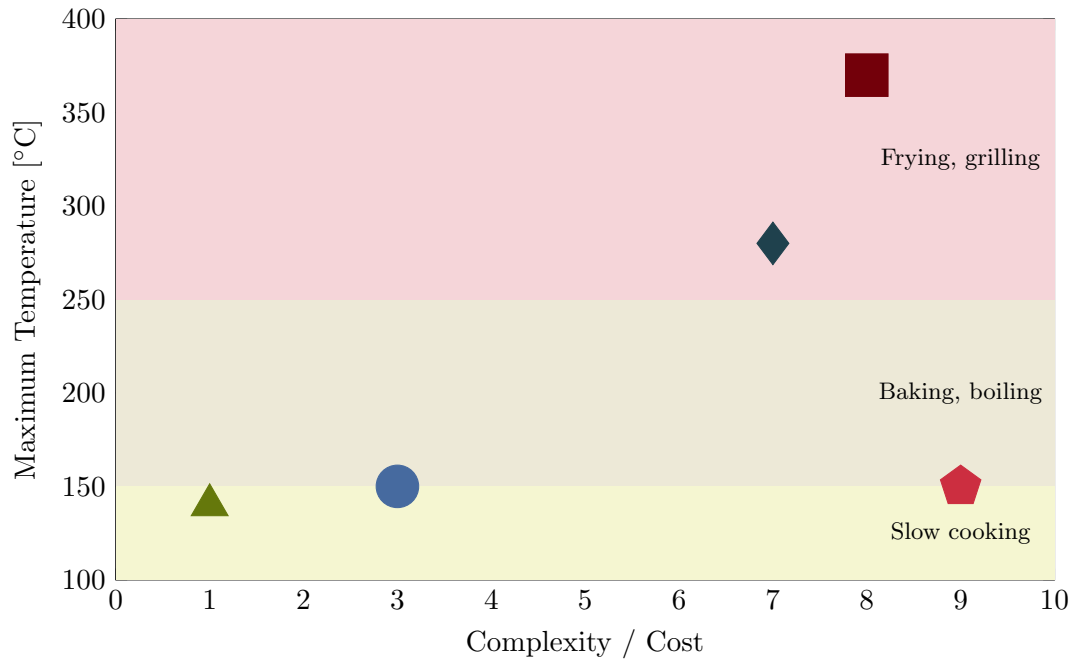


Figure 6: Solar cooker types mapped by complexity/cost vs. maximum temperature capability.

8 Recommendations by Use Case

Use Case	Recommended Type	Reason
Backpacking/emergency	Panel (CooKit)	Light, cheap, foldable
Home daily cooking	Box cooker	Reliable, no tracking, safe
High-temp cooking (frying)	Parabolic	Only option for >250°C
Cold/cloudy climates	Evacuated tube	Works in poor conditions
Evening/indoor cooking	Hybrid + PCM	Thermal storage capability
Community kitchen	Hybrid + PCM	Scalable, consistent output